JINA-CLIP-V2: MULTILINGUAL MULTIMODAL EMBEDDINGS FOR TEXT AND IMAGES

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arXiv:2412.08802v2 [cs.CL] 24 Apr 2025

ABSTRACT

Contrastive Language-Image Pretraining (CLIP) has been widely used for crossmodal information retrieval and multimodal understanding tasks. However, CLIP models are mainly optimized for crossmodal vision-language tasks and underperform in single-mode text tasks. Moreover, these models are often trained on English datasets and therefore lack multilingual understanding. Additionally, from a visual understanding perspective, previous CLIP-based models exhibit insufficient understanding of visually rich documents. In this work, we propose `jina-clip-v2`, a contrastive vision-language model trained on text pairs, triplets and image-text pairs via a multi-task and multi-stage contrastive learning paradigm in order to support both text-only and crossmodal tasks. We employ a multilingual text encoder and expand the training dataset to include multilingual texts from 29 non-English languages, including Hindi, Chinese, German, French, and others, as well as images of visually-rich documents. We evaluate the model's performance and show that `jina-clip-v2` achieves notable improvements over state-of-the-art CLIP-based models in zero-shot text-only retrieval, semantic textual similarity, and crossmodal retrieval tasks in both English and multilingual settings. `jina-clip-v2` also provides for flexibility in embedding dimensionality, enabling users to select the granularity of the representations. `jina-clip-v2` is publicly available at `https://huggingface.co/jinaai/jina-clip-v2`.

1 INTRODUCTION

Contrastive text-image pretraining is a well-known architecture and training framework for building robust text-image alignment models (Radford et al., 2021; Zhai et al., 2023; Sun et al., 2023). These models are particularly effective for tasks like crossmodal retrieval and zero-shot classification, demonstrating strong generalization capabilities. However, although CLIP models produce general-purpose embeddings for both texts and images, they are woefully inadequate as a source of text embeddings (Koukounas et al., 2024). We believe this is due to the training regimen of CLIP models, which are typically trained with short and information-poor image captions (Zhang et al., 2024a) and omit standard techniques for finetuning meaningful text embeddings, e.g. training using hard negatives (Gillick et al., 2019; Qu et al., 2021).

To overcome these limitations, Koukounas et al. (2024) introduced a multi-task, multi-stage contrastive learning approach to simultaneously align text-text and text-image embeddings. This method involved three stages: optimizing text-text and text-image embeddings for short image-caption and text-text pairs (stage 1), refining alignment using long text-text pairs and detailed image-caption pairs (stage 2), and further enhancing performance with text triplets containing hard negatives and detailed image-caption pairs (stage 3). The resulting model, `jina-clip-v1`, achieves strong performance on both the crossmodal CLIP Benchmark[1] and the text embedding MTEB Benchmark

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[1]https://github.com/LAION-AI/CLIP_benchmark

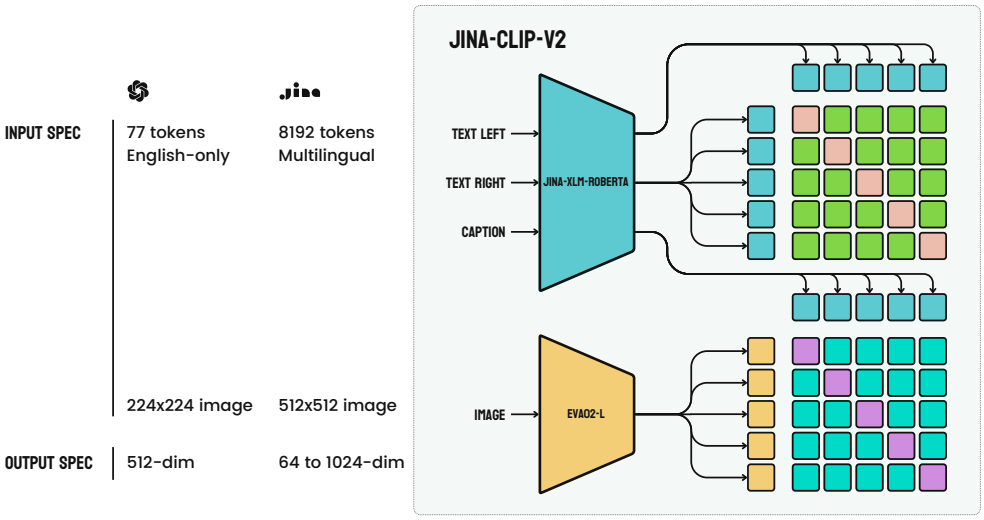

Figure 1: `jina-clip-v2` combines a text encoder (Jina XLM-RoBERTa, 561M parameters) and a vision encoder (EVA02-L14, 304M parameters) for a total of 865M parameters.

(Muennighoff et al., 2023). Despite its strengths, `jina-clip-v1` (Koukounas et al., 2024) has several limitations. First, it is an English-only model, which makes it unsuitable for multilingual document retrieval. Second, `jina-clip-v1` struggles with visually-rich images, e.g. those containing text, tables, graphs, and diagrams (Faysse et al., 2024). Moreover, `jina-clip-v1` only supports fixed length embeddings, which can be wasteful when embedding vectors are much larger than required for a task (Kusupati et al., 2024).

To address these challenges, we propose several enhancements to the pretraining scheme. For multilingual support, our approach uses a multilingual language model (Conneau et al., 2020) to initialize the text encoder, and incorporates multilingual text-image and text-text pairs during training. To improve performance on visually complex documents, we gradually increase image resolution during pretraining and use multimodal training pairs with complex visual structures. Finally, to enable users to select the granularity of their representations, we employ *Matryoshka Representation Learning* (Kusupati et al., 2024) and enable dimensional truncation on the output vectors with minimal performance degradation.

The resulting model, `jina-clip-v2` depicted in Figure 1, not only achieves competitive performance on crossmodal retrieval benchmarks against multilingual state-of-the-art models like NLLB-CLIP (Visheratin, 2023b) but also performs comparably to dedicated multilingual text embedding models like `jina-embeddings-v3` (Sturua et al., 2024) on the retrieval and semantic textual similarity (STS) tasks of the MTEB benchmark (Muennighoff et al., 2023). Moreover, due to the inclusion of visually rich training data and the progressive increase in image resolution, `jina-clip-v2` demonstrates significantly improved performance on ViDoRe, a benchmark for visually rich document retrieval (Faysse et al., 2024), compared to `jina-clip-v1`. In summary, our contributions are the following:

1. **Support for multiple languages**. `jina-clip-v2` achieves up to 67% higher scores on multilingual crossmodal performance, up to 60% on multilingual Retrieval performance and up to 43% on multilingual STS, compared `jina-clip-v1`.

2. **Improved performance on visual document retrieval**. `jina-clip-v2`'s scores are 35% higher on the ViDoRe benchmark (Faysse et al., 2024) compared to `jina-clip-v1`.

3. **Flexible embedding dimensionality**. Truncating `jina-clip-v2` embeddings by up to 75% (from 1,024 dimensions down to 256) reduces performance by $<$1%.

2 Related Work

Previous work (Jia et al., 2021; Li et al., 2021; Radford et al., 2021; Zhang et al., 2022) pioneered dual-encoder architectures trained contrastively on image-text pairs, demonstrating impressive zero-shot performance and laying the groundwork for pretrained vision-language models. Building on this foundation, Zhai et al. (2023) proposed an alternative and more efficient sigmoid objective, while Sun et al. (2023) optimized the model's dimensions and scaled up both dataset and model sizes, achieving state-of-the-art performance on crossmodal tasks. Koukounas et al. (2024) enhanced a CLIP-based retriever with strong text-to-text and crossmodal retrieval capabilities, however, their work fails to address retrieval scenarios involving candidate pools with heterogeneous modalities, due to the modality gap (Liang et al., 2022; Eslami & de Melo, 2024) and the modality bias (Lin et al., 2024). To mitigate the modality bias, Lin et al. (2024) finetuned a multimodal LLM with modality-aware hard-negative mining, creating a universal multimodal retriever and maintaining competitive text-to-text retrieval performance across diverse tasks. In this work, we build on the training strategy proposed by Koukounas et al. (2024) and train our model using a text-to-text as well as a text-to-image contrastive InfoNCE (Van den Oord et al., 2018) loss.

Extensive research has also focused on extending CLIP to languages beyond English. To address the lack of image-caption pairs in languages other than English, Carlsson et al. (2022) applied knowledge distillation to retrain the text encoder using machine-translated data. NLLB-CLIP (Visheratin, 2023a) leverages *Locked-image Tuning (LiT)* (Zhai et al., 2022) alongside the multilingual NLLB (Costa-jussà et al., 2022) text encoder, achieving state-of-the-art results in both retrieval and classification tasks.

Multilingual text embedding models are typically based on either an encoder-only architecture, such as XLM-RoBERTa (Conneau et al., 2020), or a decoder-only multilingual large language model, such as Mistral 7B (Jiang et al., 2023). Multilingual E5 (Wang et al., 2024) and BGE-M3 (Chen et al., 2024) both use XLM-RoBERTa as their backbone, the first leveraging extensive multilingual training with instruction tuning, and the second employing multi-task learning techniques. mGTE (Zhang et al., 2024b) achieves comparable performance to BGE-M3 using a smaller transformer architecture, enhanced by RoPE (Su et al., 2023) to extend the input context length. Similarly, Sturua et al. (2024) employ RoPE to increase the model's context length and apply LoRA tuning (Hu et al., 2021) to optimize embeddings for downstream tasks.

Finally, Kusupati et al. (2024) propose a training technique, which they call *Matryoshka Representation Learning*, that enables embedding models to learn coarse-to-fine representations. These representations can be truncated during inference to match the requirements of downstream tasks, cutting down on potentially redundant vector dimensions.

3 `jina-clip-v2`

The `jina-clip-v2` model uses the dual encoder architecture, introduced in the original CLIP (Radford et al., 2021) model and reused in `jina-clip-v1` (Koukounas et al., 2024).

The text encoder is initialized with pretrained Jina-XLM-RoBERTa model weights. Introduced in `jina-embeddings-v3` (Sturua et al., 2024), the Jina-XLM-RoBERTa model is a port of the multilingual XLM-RoBERTa (Conneau et al., 2020) model to a modern encoder-only architecture with Flash Attention (Dao et al., 2022), rotary positional embeddings (Su et al., 2023) and support for low-rank adaptation (Hu et al., 2021).

Like `jina-clip-v1`, the image encoder is a pretrained EVA02 model (Fang et al., 2023b). We selected the L/14 pretrained model variant, which is similar in number of parameters to the text encoder. This model implementation includes 2D rotary positional embeddings and a memory-efficient attention implementation based on *xFormers* (Lefaudeux et al., 2022). Table 1 presents architectural details for both encoders.

3.1 Training Data

Similarly to `jina-clip-v1` (Koukounas et al., 2024), we constructed four training datasets used in different stages of the training process. Dataset $\mathbb{D}^{\text{txt;p}}$ denotes a dataset of text pairs, $\mathbb{D}^{\text{txt;t}}$ a

Table 1: Model properties

Feature	Text Encoder	Image Encoder
Base Model	Jina-XLM-RoBERTa (Sturua et al., 2024)	EVA02 L/14 (Sun et al., 2023)
Parameters	561M	304M
Input Specification	8,192 tokens (max)	$(512, 512)$ resolution
Output Dimensions	1,024	1,024
Layers	24	24
Attention Implementation	FlashAttention2 (Dao et al., 2022)	xFormers (Lefaudeux et al., 2022)
Pooling Strategy	Mean pooling	CLS pooling
Additional Features	89 languages supported	Patch size 14x14

dataset of text samples with hard negatives, $\mathbb{D}^{\mathrm{mm;s}}$ a multimodal short-caption dataset and $\mathbb{D}^{\mathrm{mm;l}}$ a multimodal long-caption dataset.

Both $\mathbb{D}^{\mathrm{txt;p}}$ and $\mathbb{D}^{\mathrm{txt;t}}$ contain data in 30 languages drawn from various existing datasets for text information retrieval and semantic similarity, and were introduced as training data for `jina-embeddings-v3` (Sturua et al., 2024). $\mathbb{D}^{\mathrm{txt;p}}$ consists of pairs of texts drawn from a diverse collection of datasets, and $\mathbb{D}^{\mathrm{txt;t}}$ is a high-quality dataset with hard negatives from various sources, where each sample contains one annotated positive and seven negative items. Sturua et al. (2024) provide detailed information about these datasets and how they were curated.

Our multimodal training datasets draw on multiple sources. We randomly sampled ∼400M image-caption pairs from the ∼2B in the DFN dataset (Fang et al., 2023a) to obtain a collections of image-text pairs in English. To this, we added non-English image-text pairs sampled from CommonPool (Gadre et al., 2023). We filtered this data by language, aspect ratio, and by removing any caption with less than five words. We used multilingual SigLIP (Zhai et al., 2023) to rank image-text pairs based on cosine similarity and kept the top scoring ∼50%. The result was an additional ∼400M image-text pairs. All images were resized to $(384, 384)$. These two large-scale datasets are included in $\mathbb{D}^{\mathrm{mm;s}}$ and a small part with captions longer than 256 tokens is held out for $\mathbb{D}^{\mathrm{mm;l}}$.

In order to overcome the limitations of visual document understanding, we diversify our training data with PDFs, scientific graphs, infographics and Wikipedia images. Specifically, the short caption image-text dataset $\mathbb{D}^{\mathrm{mm;s}}$ and the long-caption $\mathbb{D}^{\mathrm{mm;l}}$ include the following training datasets: DocVQA (Mathew et al., 2021b), TatDQA (Zhu et al., 2022), InfographicsVQA (Mathew et al., 2021a), SciGraphQA (Li & Tajbakhsh, 2023), ArXivQA and ArXivCAP (Li et al., 2024), WIT (Srinivasan et al., 2021) and ViDoRe synthetic training data (Faysse et al., 2024). For the QA datasets, we concatenated the query and answer for each sample to construct a matching text in lieu of a caption. For the WIT dataset, we use the reference caption as corresponding text in $\mathbb{D}^{\mathrm{mm;s}}$ and a concatenation of reference caption, page title, section title, page description and section description as matching text for $\mathbb{D}^{\mathrm{mm;l}}$.

Finally, $\mathbb{D}^{\mathrm{mm;l}}$ includes multilingual synthetic long captions, similar to how Koukounas et al. (2024) make use of ShareGPT4v (Chen et al., 2023a). We use the GPT4v API (OpenAI, 2023) to generate detailed image descriptions for 40,000 images in 30 languages. This adds ∼1.2M generated multilingual long captions to $\mathbb{D}^{\mathrm{mm;l}}$.

3.2 Training Stages

Inspired by Koukounas et al. (2024), we employ a multi-task, multi-stage training approach, in order to optimize the model for two tasks simultaneously: text-image matching and text-text matching.

Stage 1 focuses on aligning the multimodal representations while also improving the text representations of the text encoder. We train on $\mathbb{D}^{\mathrm{txt;p}}$ and $\mathbb{D}^{\mathrm{mm;s}}$, with a small context length of 77 and an image resolution of $(224, 224)$ to enable large batch sizes of $16,384$. This is the most computationally expensive stage in our training pipeline, requiring approximately 10 days on 8 NVIDIA H100 GPUs to converge. Towards the end, when performance peaks, we switch to a higher image resolution of $(384, 384)$ using positional embedding interpolation as a warm-up for the next stage.

Stage 2 trains on $\mathbb{D}^{\text{txt;p}}$ and $\mathbb{D}^{\text{mm;l}}$, with context length increased from 77 to 512 and image resolution set at $(384, 384)$. This stage improves text embedding performance on longer text lengths, while maintaining alignment between the two modalities.

Stage 3 uses hard negatives from $\mathbb{D}^{\text{txt;t}}$ to further improve the text encoder in distinguishing relevant from irrelevant text. To maintain text-image alignment, we continue training with $\mathbb{D}^{\text{mm;l}}$. We do one more positional embedding interpolation to increase the image resolution to $(512, 512)$ and train on this resolution for the duration of stage 3.

Table 6 in Section A specifies the training details for each step.

3.3 Loss Functions

Given a batch $\mathbf{B} \subset \mathbb{D}^{\text{pairs}}$ of embedding pairs $(\mathbf{q}, \mathbf{p})$, the InfoNCE loss $\mathcal{L}_{\text{nce}}$ (Van den Oord et al., 2018), given in equation 1, evaluates the cosine similarity $cos(\mathbf{q}, \mathbf{p})$ between query $\mathbf{q} \in \mathbb{R}^{\mathbf{d}}$ and its corresponding target $\mathbf{p} \in \mathbb{R}^{\mathbf{d}}$, relative to the similarity of all other targets in the batch. The loss is calculated in both directions to preserve the symmetry of similarity measures. The temperature parameter τ influences how the loss function weighs minor differences in the similarity scores.

$$\mathcal{L}_{\text{nce}}(\mathbf{B}) := \mathcal{L}_{\text{nce}}^{\longrightarrow}(\mathbf{B}) + \mathcal{L}_{\text{nce}}^{\longleftarrow}(\mathbf{B}) \quad \text{where,}$$

$$\mathcal{L}_{\text{nce}}^{\longrightarrow}(\mathbf{B}) := \mathbb{E}_{(\mathbf{q},\mathbf{p})\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{q},\mathbf{p})/\tau}}{\sum_{i=1}^{k} e^{cos(\mathbf{q},\mathbf{p_i})/\tau}}\right], \mathcal{L}_{\text{nce}}^{\longleftarrow}(\mathbf{B}) := \mathbb{E}_{(\mathbf{q},\mathbf{p})\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{p},\mathbf{q})/\tau}}{\sum_{i=1}^{k} e^{cos(\mathbf{p},\mathbf{q_i})/\tau}}\right] \tag{1}$$

When hard negatives are available for query q, an extended version of the $\mathcal{L}_{\text{nce}}$ loss is used. Given a batch $\mathbf{B} \subset \mathbb{D}^{\text{triplets}}$ of samples $r = (\mathbf{q}, \mathbf{p}, \mathbf{n_1}..., \mathbf{n_7})$, consisting of a query $\mathbf{q}$, a positive match $\mathbf{p}$, and seven negatives $\mathbf{n_1}..., \mathbf{n_7}$, the extended loss function, denoted here as $\mathcal{L}_{\text{nce}^+}$, is given in equation 2. Similarly to $\mathcal{L}_{\text{nce}}$, this loss function is bidirectional.

$$\mathcal{L}_{\text{nce}^+}(\mathbf{B}) := \mathbb{E}_{r\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{q},\mathbf{p})/\tau}}{\sum_{i=1}^{k}\left[e^{cos(\mathbf{q},\mathbf{p_i})/\tau} + \sum_{j=1}^{7} e^{cos(\mathbf{q},\mathbf{n_{j,i}})/\tau}\right]}\right] + \mathbb{E}_{r\sim\mathbf{B}}\left[-\ln \frac{e^{cos(\mathbf{p},\mathbf{q})/\tau}}{\sum_{i=1}^{k} e^{cos(\mathbf{p},\mathbf{q_i})/\tau}}\right] \tag{2}$$

Equation 3 defines the loss for each of the three stages. In each stage, we optimize a joint loss function, i.e. the sum of two loss functions, one operating on text representations and one on multimodal representations. In equation 3, $\mathbf{B}_k$ denotes a batch of samples drawn from the dataset $\mathbb{D}^{\text{k}}$, where $\text{k} \in \{txt; p, txt; t, mm; s, mm; l\}$. All stages optimize $\mathcal{L}_{\text{nce}}$ except Stage 3, which uses hard negatives in the text loss branch and thus calculates $\mathcal{L}_{\text{nce}^+}$.

$$\begin{aligned}
\mathcal{L}_1(\mathbf{B}_{txt;p}, \mathbf{B}_{mm;s}) &:= \mathcal{L}_{\text{nce}}(\mathbf{B}_{txt;p}) + \mathcal{L}_{\text{nce}}(\mathbf{B}_{mm;s}) \\
\mathcal{L}_2(\mathbf{B}_{txt;p}, \mathbf{B}_{mm;l}) &:= \mathcal{L}_{\text{nce}}(\mathbf{B}_{txt;p}) + \mathcal{L}_{\text{nce}}(\mathbf{B}_{mm;l}) \\
\mathcal{L}_3(\mathbf{B}_{txt;t}, \mathbf{B}_{mm;l}) &:= \mathcal{L}_{\text{nce}^+}(\mathbf{B}_{txt;t}) + \mathcal{L}_{\text{nce}}(\mathbf{B}_{mm;l})
\end{aligned} \tag{3}$$

3.4 Matryoshka Representation Learning

In every training stage, the loss component is recalculated at different dimensionalities: Loss is computed for the full 1024-dim output vector, and for truncated subsets of 64, 128, 256, 512 and 768 dimensions. Model weight adjustment proceeds on the basis of all those losses combined. This training technique, called *Matryoshka Representation Learning* (Kusupati et al., 2024), induces the model to encode its embeddings by increasing granularity. This makes it possible, at inference time, to truncate embedding vectors with a minor performance penalty.

4 Evaluation

We evaluate the performance of `jina-clip-v2` on a selection of benchmarks: English and multilingual crossmodal retrieval benchmarks in Section 4.1, text embedding tasks from the MTEB

Table 2: Evaluation results on crossmodal retrieval tasks

Benchmark		**CLIP Benchmark**		**Crossmodal-3600**		**XTD10**	
Language		English		Multilingual		Multilingual	
Task Type		Zero-Shot Retrieval					
Model	**#Parameters**	T-I r@5	I-T r@5	T-I r@5	I-T r@5	T-I r@5	I-T r@5
`jina-clip-v1`	223M	77.75	87.65	16.93	19.82	31.01	36.89
nllb-siglip-base	507M	79.57	86.38	79.29	76.56	86.23	84.87
nllb-siglip-large	1.2B	**81.54**	88.15	82.07	80.16	**87.60**	85.37
`jina-clip-v2`stage 1	865M	73.87	86.61	73.51	79.64	80.66	83.02
`jina-clip-v2`stage 2	865M	79.81	89.57	**84.13**	**84.12**	86.11	**86.45**
`jina-clip-v2`	865M	79.09	**89.73**	81.43	83.23	84.87	86.03

T-I r@5: Text to Image Recall@5 [%] I-T r@5: Image to Text Recall@5 [%]

benchmark suite in Section 4.2 and visual document retrieval tasks in Section 4.3. Finally, we present an ablation study of `jina-clip-v2`'s Matryoshka representations in Section 4.4. Detailed evaluation results are presented in Section A tables 7 to 26.

4.1 Crossmodal Retrieval Evaluation

We evaluate `jina-clip-v2` on a set of English and multilingual crossmodal tasks, conducting a comparative analysis with both `jina-clip-v1` and the state-of-the-art NLLB-CLIP (Visheratin, 2023b) large and base SigLIP variants. Additionally, we track the model's performance at all training stages to evaluate the effectiveness of the training protocol. For English zero-shot image-text and text-image retrieval, we conduct evaluations on the CLIP Benchmark, which includes Flickr30K (Young et al., 2014) and COCO Captions (Chen et al., 2015). For multilingual crossmodal retrieval tasks, we assess performance on the Crossmodal-3600 (Thapliyal et al., 2022) and XTD10 (Aggarwal & Kale, 2020; Rajendran et al., 2016) datasets.

Table 2 demonstrates the strong performance of `jina-clip-v2` across both English and multilingual benchmarks. On the English CLIP Benchmark, our model outperforms `jina-clip-v1` and while `nllb-siglip-large` leads in text-to-image retrieval, `jina-clip-v2` outperforms both NLLB-CLIP variants in image-to-text retrieval. On multilingual multimodal retrieval, we obtain competitive results, approaching the performance of `nllb-siglip-large` in text-to-image retrieval and surpassing it in image-to-text retrieval. Regarding stage progression, performance improves considerably between Stage 1 and Stage 2 on all benchmarks, but drops slightly from Stage 2 to Stage 3. We hypothesize that there is a trade-off between crossmodal and text-only retrieval performance, and during stage 3 training, where focus shifts towards text embedding training with hard negatives, the model falls slightly behind on crossmodal retrieval.

4.2 Text Retrieval and STS Evaluation

Table 3 presents the results of evaluations on the retrieval and semantic textual similarity (STS) tasks from MTEB (Muennighoff et al., 2023), both in English-only and multilingual contexts. `jina-clip-v2` significantly improves on the performance of `nllb-siglip-large` in both English and multilingual text tasks, demonstrating the value of explicit text retrieval training. Compared to `jina-clip-v1`, performance is significantly better on multilingual tasks, and comparable on English-only tasks. Compared to the text-only frontier model `jina-embeddings-v3`, performance is lackluster on retrieval tasks. This suggests that multimodal training is hampering text-only performance.

4.3 Visual Document Retrieval

Visual document retrieval challenges vision-language models to capture fine-grained information from images, including embedded text, and accurately match these with text queries or documents. This task is particularly challenging as it deals with high-resolution document images with complex content, which differs significantly from processing typical image-caption

Table 3: Evaluation results on MTEB (Muennighoff et al., 2023) Retrieval and STS tasks across 8 languages

Retrieval nDCG@10 [%]									
Model	**#Parameters***	en	zh	hi	de	fr	es	jp	ru
jina-clip-v1	137M	47.04	8.09	8.73	33.77	39.99	39.77	10.77	4.78
nllb-siglip-base	414M	24.69	27.04	39.18	22.67	28.30	30.06	30.73	32.68
nllb-siglip-large	767M	27.83	32.96	43.22	32.29	38.74	39.92	35.53	38.87
jina-embeddings-v3	572M	**48.34**	**64.27**	**65.18**	**63.59**	**64.09**	**62.97**	**68.47**	**66.46**
jina-clip-v2stage 1	561M	42.02	55.11	62.03	57.56	58.56	59.86	63.34	60.74
jina-clip-v2stage 2	561M	43.47	57.66	62.19	59.16	61.05	61.51	65.13	63.66
jina-clip-v2	561M	46.46	60.47	63.08	56.66	61.16	61.48	65.48	64.73
STS Spearman correlation based on cosine similarity									
Model	**#Parameters***	en	zh	hi	de	fr	es	jp	ru
jina-clip-v1	137M	81.35	22.97	36.09	50.04	67.97	70.05	55.83	39.77
nllb-siglip-base	414M	72.57	33.36	68.11	39.58	63.94	65.28	77.39	57.55
nllb-siglip-large	767M	74.73	39.15	73.78	53.45	72.38	69.68	79.25	62.84
jina-embeddings-v3	572M	**81.68**	54.60	83.62	75.32	81.11	80.78	81.03	77.16
jina-clip-v2stage 1	561M	80.01	52.74	83.35	74.84	79.06	80.42	81.48	75.62
jina-clip-v2stage 2	561M	80.76	53.37	**84.30**	75.64	80.19	80.51	81.46	76.18
jina-clip-v2	561M	81.58	**55.08**	79.55	**76.27**	**81.70**	**81.89**	**82.01**	**77.79**

* Refers to the parameters of the text tower

Table 4: ViDoRe benchmark (Faysse et al., 2024) evaluation results

Task	jina-clip-v2	jina-clip-v2 stage 1	jina-clip-v2 stage 2	jina-clip-v1	nllb-siglip large	siglip-so400m*
Retrieval - nDCG@5 [%]						
ArxivQ	**64.92**	58.78	66.14	25.40	30.44	43.2
DocQ	24.64	21.16	24.81	11.90	23.82	**30.3**
InfoQ	57.90	56.62	56.63	35.50	59.87	**64.1**
TabF	45.91	37.28	43.18	20.20	**70.68**	58.1
TATQ	**30.25**	12.09	25.79	3.30	20.00	26.2
Shift	**34.07**	28.41	31.54	3.80	30.79	18.7
AI	**68.07**	32.95	55.64	15.20	47.90	62.5
Energy	62.15	49.15	60.97	19.70	64.94	**65.7**
Gov	**68.97**	37.81	55.56	21.40	58.62	66.1
Health	69.05	39.50	57.37	20.80	58.43	**79.1**
Average	**52.65**	37.37	47.76	17.72	46.55	51.4

* (Zhai et al., 2023; Alabdulmohsin et al., 2024), scores taken directly from Faysse et al. (2024)

datasets. As shown in Table 4, jina-clip-v2 outperforms other state-of-the-art CLIP models on the ViDoRe Benchmark for visual document understanding (Faysse et al., 2024), with an average nDCG@5 score of 52.65%, toping jina-clip-v1, nllb-siglip-large and siglip-so400m-patch14-384 (Zhai et al., 2023; Alabdulmohsin et al., 2024). The progression through training stages brings substantial improvements, highlighting the effectiveness of gradually increasing the image resolution. We further investigate how image resolution affects visual document retrieval in Section 5.1.

4.4 Matryoshka Representation Learning

Table 5 presents the impact of embedding truncation on the crossmodal and text-only tasks. Performance remains highly stable when reducing dimensions from 1024 to 256, with minimal degradation (typically less than 1%) across all evaluation tasks. At 256 dimensions, representing a 75% reduction

Table 5: MRL (Kusupati et al., 2024) ablation study on various embedding dimensions

Dataset - Dimension	1024	768	512	256	128	64
Text-Image Retrieval - Recall@5 [%]						
CLIP Benchmark	79.10	**79.12**	78.93	78.32	75.90	70.51
Crossmodal-3600 (Thapliyal et al., 2022)	81.43	**82.35**	82.31	81.75	78.17	72.52
XTD10 (Aggarwal & Kale, 2020; Rajendran et al., 2016)	**84.87**	84.85	84.60	84.32	81.80	77.85
Image-Text Retrieval - Recall@5 [%]						
CLIP Benchmark	**89.73**	89.60	89.55	89.35	87.48	83.20
Crossmodal-3600 (Thapliyal et al., 2022)	83.23	**83.26**	83.21	82.81	80.54	75.37
XTD10 (Aggarwal & Kale, 2020; Rajendran et al., 2016)	**86.03**	86.02	86.02	85.84	84.37	81.02
Text-Text Retrieval - nDCG@10 [%]						
EN Classic MTEB (Muennighoff et al., 2023) Retrieval	**49.33**	49.32	49.19	48.67	46.37	40.66
Semantic Textual Similarity - Spearman Correlation [%]						
EN Classic MTEB (Muennighoff et al., 2023) STS	**81.29**	81.27	81.26	81.24	80.78	79.56

in embedding size, the model preserves effectively all essential semantic information. Substantial performance degradation is only observed at dimensions 128 and 64.

5 Analysis

5.1 The Role of Image Resolution in Visual Document Retrieval

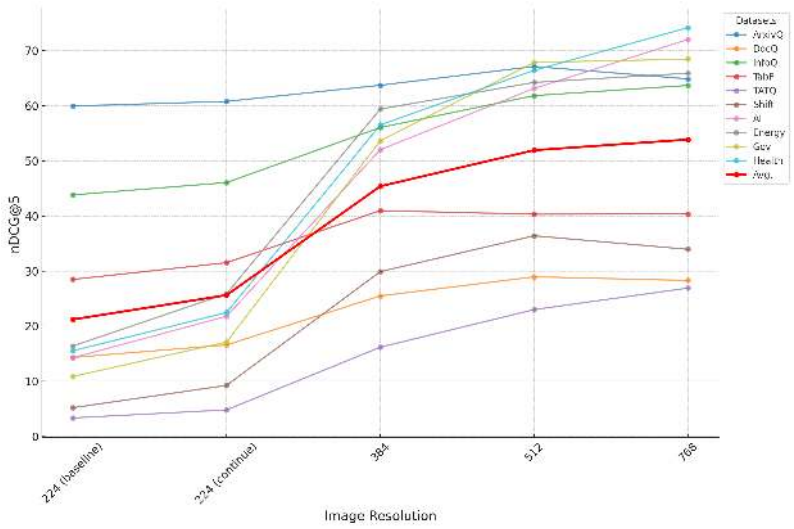

Figure 2: Performance on the ViDoRe benchmark (Faysse et al., 2024) against input resolution

Our analysis of performance at various stages of training demonstrates that image resolution plays a critical role in retrieving visually rich documents. To better understand the relationship between image resolution and model performance on such documents, we conduct a targeted experiment.

We checkpointed `jina-clip-v2` at the end of Stage 1, when it had only seen images with a resolution of $(224, 224)$. We conducted four runs, keeping the image resolution to $(224, 224)$ for the first run and increasing to $(384, 384)$, $(512, 512)$, and $(768, 768)$ for three additional runs. Each run was trained for 3500 steps using the same visually rich training set and hyperparameters. The models were then evaluated on the ViDoRe benchmark (Faysse et al., 2024), which comprises 10

datasets designed for retrieving visually rich documents with text queries. We plot the results in Figure 2.

Unsurprisingly, increasing image resolution has a positive impact on linking queries to visually rich documents. The most significant improvement occurs when resolution increases from $(224, 224)$ to $(384, 384)$, with the average nDCG@5 score across 10 benchmarks rising from 0.256 to 0.454. A further increase in resolution to $(512, 512)$ also brings a noticeable gain in performance.

Considering that the model is trained with a patch size of 14, increasing the resolution beyond $(512, 512)$ raises the number of patches quadratically, for example, by a factor of 2.25x when increasing the resolution to $(768, 768)$. This large increase in computing costs yields only marginal improvements, with nDCG@5 increasing by just 0.019. We consider $(512, 512)$ the optimal image resolution for this model, with a reasonable balance between performance and cost.

5.2 Unified Batch Training vs Multi-Task Learning

Our training approach, as outlined in Section 3, involves two tasks at each stage, both optimized in a multi-task learning framework: a multimodal alignment task for images and text, and a second text-only task. Both use the InfoNCE (Van den Oord et al., 2018) objective. Regardless of whether the loss is computed on image-text or text-text pairs, the same loss function is applied to pairs of vectors. This raises the question: Is there any value in merging the image-text pair and text-text pair data into mixed modality batches and optimizing a single contrastive objective?

We refer to this approach as the *Unified Batch Training* technique, in contrast to the *Multi-Task Learning* paradigm. The motivation is two-fold: simplicity of computation, and the possibility of mitigating the modality gap (Liang et al., 2022) by using in-batch negatives across modalities. Compared to multi-task training, the unified batch approach would use a single InfoNCE loss with a shared temperature value, attempting to force all modalities to align within the same embedding space. For this investigation, we also integrated a SimCLR-like self-supervised learning method (Chen et al., 2020) into the contrastive training, aiming to provide more robust image representations. A schematic representation of the unified batch technique is given in Figure 3 in Section A.

Our Stage 1 experiments show that this technique does bring some early improvements, but fails beyond a certain point, at least with regard to crossmodal tasks. This was reflected in the slow decay of the loss temperature τ (our temperature is a trainable parameter in this case) and its eventual stabilization around 0.02. This is in contrast to the multi-task learning approach, where the temperature value decreases more rapidly and keeps decreasing throughout training down to 0.015, leading to better performance on crossmodal tasks.

We hypothesize that this limitation arises from the fundamental information asymmetry between the visual and textual modalities (Schrodi et al., 2024). Images are very information-dense, full of small details, while the corresponding textual descriptions are far less detailed (Schrodi et al., 2024; Chen et al., 2023b). In temperature-scaled contrastive learning, the temperature parameter controls the hardness of the loss function with respect to the negative samples, i.e., the level of penalties on the hard negative samples (Wang & Liu, 2021). From an information-theory perspective, the level of penalty on the hard negative text-text samples theoretically differs from that of image-text samples due to the information gap problem. Consequently, as supported by our experiments, enforcing a unified temperature parameter across both modalities is empirically suboptimal.

6 Conclusion

This work presents an enhanced strategy for training dual-encoder vision-language embedding models for multilingual crossmodal as well as text retrieval and semantic text similarity tasks using contrastive learning. We introduce improvements to the training, namely support for multiple languages, Matryoshka Representation Learning, and visually-rich document understanding. We have used these improvements to train and release a new model, `jina-clip-v2`, with strong crossmodal and text-only performance on standard benchmarks. Finally, we have presented in this paper analyses of two important considerations for CLIP models going forward: the effects of the modality gap and the role of the image resolution in understanding complex visual inputs.

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A APPENDIX

Table 6: Training settings on each stage

Parameter	**Stage 1**	**Stage 2**	**Stage 3**
Image encoder weights init	EVA02 ViT L/14 (Sun et al., 2023)	Stage 1	Stage 2
Text encoder weights init	JinaXLMRoBERTa (Sturua et al., 2024)	Stage 1	Stage 2
Peak image encoder LR	2e-4	5e-5	5e-6
Image encoder layer-wise LR decay	1	0.98	1
Peak text encoder LR	1e-4	5e-5	1e-4
Text encoder layer-wise LR decay	1	0.98	1
Image-text pairs batch size	$16,384$	$8,192$	$1,024$
Text pairs batch size	$16,384$	$8,192$	128
Total steps	$100,000$	$6,000$	$16,000$
Max sequence length	77	512	512
Image-text pairs samples seen	1.7B	50M	16M
Text pairs samples seen	1.7B	50M	2M
Number of GPUs - H100s 80GB	8	8	8
Input resolution	$(224, 224) \rightarrow (384, 384)$	$(384, 384)$	$(512, 512)$
Patch size	$(14, 14)$		
LR schedule	cosine decay		
Optimizer	AdamW (Loshchilov & Hutter, 2019)		
Optimizer hyper-parameters	$\beta_1, \beta_2, \epsilon = 0.9, 0.98, 1e-6$		
Weight decay	0.02		
Numerical precision	bfloat16		

Table 7: Model performance on the CLIP Benchmark retrieval tasks

Dataset - Model	jina-clip-v2	jina-clip-v2 stage 1	jina-clip-v2 stage 2	jina-clip-v1	nllb-siglip large	nllb-siglip base
Zero-shot Image Retrieval - Recall@5 [%]						
Flickr30K (Young et al., 2014)	89.84	86.84	90.04	89.04	**92.24**	90.02
MS COCO (Chen et al., 2015)	68.35	60.91	69.59	66.42	**70.84**	69.13
Zero-shot Text Retrieval - Recall@5 [%]						
Flickr30K (Young et al., 2014)	**98.00**	96.10	97.40	96.40	97.10	95.00
MS COCO (Chen et al., 2015)	81.46	77.12	**81.74**	79.02	79.20	77.76

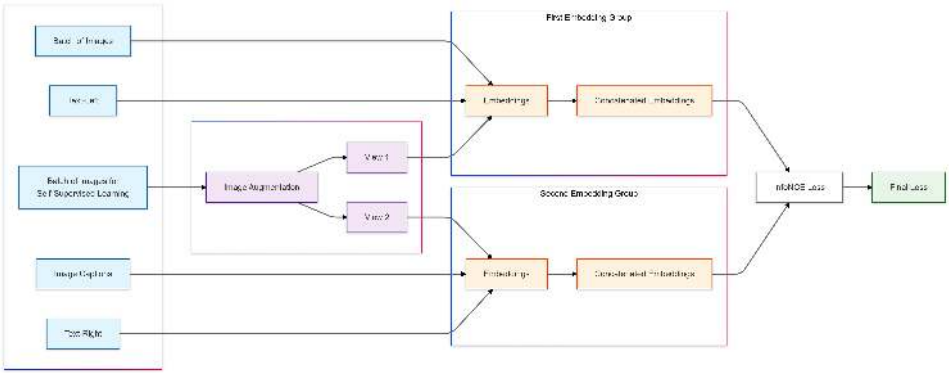

Figure 3: Contrastive learning between two embedding groups (Unified Batch technique). The first group concatenates original images, question texts, and one view of augmented images, while the second group concatenates corresponding image captions, answer texts, and a different view of the augmented images.

Table 8: Model performance on Crossmodal-3600 (Thapliyal et al., 2022)

Language - Model	jina-clip-v2	jina-clip-v2 stage 1	jina-clip-v2 stage 2	jina-clip-v1	nllb-siglip large	nllb-siglip base
Zero-shot Image Retrieval - Recall@5 [%]						
average	81.43	73.51	**84.13**	16.93	82.07	79.29
ar	73.56	66.22	76.89	0.19	**78.92**	76.94
bn	63.78	50.19	68.11	0.11	**75.19**	74.19
da	85.39	76.53	**87.97**	15.39	87.14	86.64
de	91.25	84.64	**92.42**	37.42	89.56	87.25
el	75.03	68.53	**77.92**	0.56	77.83	71.97
en	75.83	69.78	**77.78**	76.17	73.11	72.22
es	83.64	78.28	**86.28**	47.36	82.64	79.97
fi	82.83	75.28	85.89	5.03	**86.42**	81.44
fr	88.78	83.50	**90.67**	62.33	87.86	85.58
hi	55.25	42.03	59.47	0.11	**60.31**	58.08
id	84.22	73.69	**86.94**	13.14	86.31	84.17
it	88.33	81.19	**89.64**	33.39	85.94	82.67
ja	87.03	77.28	**90.03**	3.75	86.06	83.14
ko	78.81	71.81	**83.22**	0.33	78.75	75.47
nl	82.56	75.72	**84.47**	27.69	81.69	78.86
no	81.08	71.97	**83.08**	15.61	82.69	79.97
pl	84.00	79.33	**86.50**	7.39	82.72	78.61
pt	82.42	76.17	**85.19**	31.97	82.69	79.44
ro	89.36	82.92	**92.22**	17.89	90.03	86.17
ru	88.97	82.97	**91.11**	2.19	86.44	83.78
sv	78.06	71.33	**80.56**	15.22	79.33	76.17
th	81.61	70.92	**85.08**	1.89	81.14	78.83
tr	81.31	75.31	**84.53**	4.86	83.47	81.00
uk	88.56	82.28	**89.89**	0.94	85.44	81.89
vi	86.64	76.31	**89.06**	3.17	85.56	82.56
zh	78.97	66.97	**82.50**	2.06	76.56	74.64
Zero-shot Text Retrieval - Recall@5 [%]						
average	83.23	79.64	**84.12**	19.82	80.16	76.56
ar	76.25	72.14	**76.67**	0.31	75.86	73.69
bn	69.00	63.78	70.22	0.11	**75.58**	73.61
da	88.53	84.42	**88.86**	17.25	86.53	84.69
de	**92.47**	88.42	**92.47**	39.75	87.50	84.44
el	73.33	73.22	**75.61**	0.64	74.81	68.89
en	78.58	74.33	**79.61**	79.06	70.81	69.08
es	86.28	81.19	**87.36**	49.14	81.19	77.00
fi	84.19	81.17	**85.64**	7.22	84.25	78.75
fr	90.89	86.56	**91.00**	63.14	86.64	83.64
hi	61.64	54.39	61.56	0.08	**61.89**	59.83
id	86.31	83.64	**87.64**	16.75	84.33	81.89
it	90.17	84.31	**90.53**	36.53	83.50	78.64
ja	88.50	85.53	**89.44**	7.69	84.03	81.14
ko	81.42	79.22	**83.00**	0.53	76.75	74.17
nl	82.47	77.94	**83.33**	29.28	79.28	74.33
no	83.75	77.22	**84.42**	18.53	82.08	77.03
pl	84.61	83.72	**85.69**	10.00	79.58	75.39
pt	83.94	79.31	**84.06**	33.33	79.39	74.89
ro	91.31	87.39	**92.08**	18.83	88.83	83.58
ru	90.64	87.75	**90.89**	3.33	84.64	80.61
sv	78.28	77.53	**80.11**	16.50	76.58	72.31
th	81.94	80.22	**84.11**	2.17	79.56	76.28
tr	82.67	80.36	**83.81**	6.94	80.36	77.69
uk	89.53	86.42	**89.97**	1.58	83.22	78.67
vi	88.06	85.64	**88.58**	6.06	83.14	80.03
zh	79.22	74.94	**80.42**	4.67	73.83	70.22

Table 9: Model performance on XTD10 (Aggarwal & Kale, 2020; Rajendran et al., 2016)

Language - Model	jina-clip-v2	jina-clip-v2 stage 1	jina-clip-v2 stage 2	jina-clip-v1	nllb-siglip large	nllb-siglip base
Zero-shot Image Retrieval - Recall@5 [%]						
average	84.87	80.66	86.11	31.01	**87.60**	86.23
de	85.70	80.40	86.80	48.80	**88.30**	87.00
en	**89.40**	84.00	**89.40**	89.00	**89.40**	88.80
es	85.90	82.70	87.40	56.80	**88.20**	86.70
fr	85.10	80.90	87.30	66.80	**87.70**	86.90
it	85.80	83.20	87.10	45.00	**89.30**	87.80
ko	82.10	78.30	83.00	1.30	**85.20**	83.60
pl	86.50	81.10	88.00	11.00	**89.40**	87.60
ru	81.10	79.30	82.10	3.20	**83.40**	82.40
tr	83.70	81.30	86.30	7.00	**88.30**	86.80
zh	83.40	75.40	83.70	4.50	**86.80**	84.70
Zero-shot Text Retrieval - Recall@5 [%]						
average	86.03	83.02	**86.45**	36.89	85.37	84.56
de	86.20	82.90	**86.90**	50.10	84.40	84.30
en	90.50	87.80	**90.60**	89.40	88.30	87.50
es	87.00	84.90	**87.50**	55.60	85.80	85.20
fr	85.20	82.20	85.00	67.30	**86.30**	84.40
it	**88.00**	83.10	87.70	51.10	86.70	85.60
ko	82.10	79.30	**84.00**	1.60	83.20	82.60
pl	88.50	84.50	**88.90**	15.50	86.80	86.60
ru	**83.10**	79.60	81.80	3.50	80.90	81.20
tr	85.60	83.20	85.90	9.50	**87.10**	85.40
zh	84.10	82.70	**86.20**	7.40	84.20	82.80

Table 10: Retrieval and STS evaluation on English MTEB (Muennighoff et al., 2023) (1/3)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
		Retrieval nDCG@10 [%]						
AILACasedocs	test	32.79	6.62	8.63	**34.73**	29.21	31.88	23.10
AILAStatutes	test	14.45	15.04	13.58	**33.00**	22.58	26.45	8.86
ARCChallenge	test	10.52	5.85	6.78	10.12	**11.18**	10.92	10.96
AlphaNLI	test	**31.45**	16.02	19.25	30.16	28.39	27.87	29.12
ArguAna	test	**49.41**	25.01	33.81	43.28	42.85	47.99	43.65
BelebeleRetrieval	test	91.83	72.88	77.43	**93.25**	91.84	91.86	92.97
CQADupstack AndroidRetrieval	test	52.14	27.46	31.30	37.01	52.14	**52.58**	50.21
CQADupstack EnglishRetrieval	test	45.74	19.98	23.40	44.50	**48.67**	48.19	46.36
CQADupstack GamingRetrieval	test	**59.61**	34.23	37.64	47.53	58.56	58.39	54.40
CQADupstack GisRetrieval	test	39.30	17.70	20.62	35.66	38.72	39.48	**40.61**
CQADupstack MathematicaRetrieval	test	28.23	13.08	15.66	30.23	30.69	**31.37**	30.70
CQADupstack PhysicsRetrieval	test	46.31	25.37	28.74	42.18	45.62	**46.61**	45.84
CQADupstack ProgrammersRetrieval	test	41.40	21.06	25.66	37.33	41.32	**42.61**	41.30
CQADupstackRetrieval	test	40.93	20.74	23.54	36.02	41.48	**42.17**	41.19
CQADupstack StatsRetrieval	test	33.97	19.10	20.63	31.20	35.06	**35.43**	35.41
CQADupstack TexRetrieval	test	29.43	12.42	13.46	26.97	29.71	30.41	**30.92**
CQADupstack UnixRetrieval	test	41.13	19.57	20.99	37.22	42.60	**43.01**	42.48
CQADupstack WebmastersRetrieval	test	41.43	23.31	26.58	33.16	41.81	**43.94**	42.01
CQADupstack WordpressRetrieval	test	32.43	15.65	17.87	29.20	32.92	34.05	**34.05**
CUREv1	dentistry_and_oral_health	45.53	20.08	26.11	49.19	41.41	40.88	**50.26**
CUREv1	dermatology	51.32	17.25	22.31	53.54	41.41	42.34	**59.59**
CUREv1	gastroenterology	45.20	19.99	23.25	49.36	40.56	41.59	**51.96**
CUREv1	genetics	48.26	18.91	27.19	**56.64**	41.62	43.95	54.23
CUREv1	neuroscience_and_neurology	38.67	15.97	21.31	**44.91**	35.35	35.65	42.92
CUREv1	orthopedic_surgery	45.13	12.79	19.48	44.11	40.72	38.91	**46.29**
CUREv1	otorhinolaryngology	39.65	12.28	18.19	41.91	34.29	32.90	**46.59**
CUREv1	plastic_surgery	46.10	15.43	23.43	46.86	40.10	40.62	**48.34**
CUREv1	psychiatry_and_psychology	46.87	21.02	27.88	**52.34**	41.78	42.10	50.63
CUREv1	pulmonology	45.08	16.75	26.42	**53.72**	41.09	40.11	51.80
CUREv1	avg	45.18	17.05	23.56	49.26	39.83	39.91	**50.26**
ChemHotpotQA Retrieval	test	**78.05**	41.81	44.20	68.65	63.21	61.77	58.96
ChemNQRetrieval	test	57.11	39.22	39.19	**60.80**	48.21	47.78	54.58
ClimateFEVER	test	24.82	13.06	16.44	**42.26**	19.31	23.70	30.14
ClimateFEVER HardNegatives	test	25.32	14.03	17.11	**43.02**	20.16	24.22	30.47
CodeFeedbackMT	test	39.15	4.86	6.66	**59.84**	50.42	52.46	47.11
CodeFeedbackST	test	66.06	17.09	20.64	**78.09**	70.81	71.05	68.62
DBPedia	test	36.66	17.76	21.37	**41.00**	22.03	24.55	37.42
DBPediaHardNegatives	test	38.84	23.32	26.66	**43.64**	26.47	28.54	40.76
FEVER	test	76.30	29.84	33.91	**89.07**	60.54	67.09	84.98
FEVERHardNegatives	test	77.02	38.27	40.69	**89.86**	61.52	68.72	86.56
FaithDial	test	23.18	16.83	18.69	**26.35**	24.03	23.93	22.84
FeedbackQARetrieval	test	49.15	22.69	33.43	**52.06**	45.38	49.90	51.00
FiQA2018	test	38.27	8.74	12.79	**47.46**	41.65	40.96	42.00
HagridRetrieval	dev	**98.69**	96.85	97.16	98.69	98.65	98.42	98.69
HellaSwag	test	27.30	14.33	18.39	29.26	**29.32**	29.02	27.21
HotpotQA	test	61.89	21.55	23.09	**64.65**	49.63	46.96	60.15
HotpotQA HardNegatives	test	62.58	27.56	28.59	**64.74**	52.00	49.67	60.01
LEMBNarrative QARetrieval	test	33.55	10.82	12.62	**34.22**	19.86	19.68	23.57
LEMBNeedleRetrieval	test_256	72.00	36.00	40.00	64.00	78.00	72.00	**86.00**
LEMBNeedleRetrieval	test_512	58.00	24.00	18.00	32.00	64.00	54.00	**70.00**
LEMBNeedleRetrieval	test_1024	**66.00**	6.00	6.00	14.00	64.00	50.00	46.00
LEMBNeedleRetrieval	test_2048	**76.00**	8.00	8.00	10.00	60.00	56.00	58.00
LEMBNeedleRetrieval	test_4096	**66.00**	2.00	2.00	8.00	48.00	48.00	54.00
LEMBNeedleRetrieval	test_8192	**72.00**	0.00	2.00	12.00	46.00	46.00	34.00
LEMBNeedleRetrieval	test_16384	**42.00**	0.00	0.00	8.00	28.00	26.00	28.00
LEMBNeedleRetrieval	test_32768	**16.00**	2.00	4.00	2.00	6.00	8.00	8.00
LEMBNeedleRetrieval	avg	**58.50**	9.75	10.00	18.75	49.25	45.00	48.00
LEMBPasskeyRetrieval	test_256	90.00	22.00	24.00	**100.00**	98.00	76.00	100.00
LEMBPasskeyRetrieval	test_512	52.00	20.00	16.00	**100.00**	58.00	72.00	98.00
LEMBPasskeyRetrieval	test_1024	34.00	14.00	14.00	**94.00**	16.00	42.00	56.00

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 11: Retrieval and STS evaluation on English MTEB (Muennighoff et al., 2023) (2/3)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
LEMBPasskeyRetrieval	test_2048	34.00	4.00	4.00	**72.00**	12.00	14.00	12.00
LEMBPasskeyRetrieval	test_4096	**72.00**	0.00	0.00	42.00	2.00	20.00	0.00
LEMBPasskeyRetrieval	test_8192	**82.00**	0.00	2.00	26.00	10.00	4.00	8.00
LEMBPasskeyRetrieval	test_16384	**44.00**	2.00	2.00	26.00	8.00	4.00	6.00
LEMBPasskeyRetrieval	test_32768	**26.00**	2.00	0.00	10.00	6.00	8.00	4.00
LEMBPasskeyRetrieval	avg	54.25	8.00	7.75	**58.75**	26.25	30.00	35.50
LEMBQMSumRetrieval	test	37.38	8.05	9.57	**39.35**	29.37	29.93	31.07
LEMBSummScreen FDRetrieval	validation	**93.37**	23.44	23.31	92.22	55.86	56.84	75.02
LEMBWikimQARetrieval	test	**75.19**	33.00	35.49	65.95	57.39	55.88	64.85
LegalBench ConsumerContractsQA	test	69.55	39.91	44.37	**78.14**	61.34	67.18	64.18
LegalBench CorporateLobbying	test	89.09	81.55	86.17	**93.65**	88.40	90.37	91.57
LegalSummarization	test	**63.23**	45.25	50.57	59.25	53.08	56.06	62.70
LitSearchRetrieval	test	41.89	11.96	12.94	48.18	39.51	39.15	**48.37**
MIRACLRetrieval HardNegatives	dev	43.34	19.57	20.44	**51.97**	31.97	39.01	50.77
MLQARetrieval	test	63.42	35.82	40.61	**64.72**	59.87	60.03	64.08
MLQuestions	test	59.43	27.79	30.23	**63.10**	54.91	54.74	56.11
MSMARCO	dev	36.92	10.63	13.80	**40.84**	22.94	27.12	37.40
MSMARCOHardNegatives	test	65.43	41.81	44.71	**71.74**	55.21	58.27	69.89
MedicalQARetrieval	test	67.24	34.04	39.22	70.23	66.01	68.24	**71.34**
MultiLongDoc Retrieval	test	**34.78**	8.38	10.80	28.87	19.50	18.97	20.16
NFCorpus	test	33.52	16.77	22.13	**36.61**	31.00	32.16	32.88
NQ	test	58.11	16.28	19.94	**64.33**	40.00	43.03	57.16
NQHardNegatives	test	59.43	21.05	23.33	**65.06**	41.42	44.50	58.50
NanoArguAnaRetrieval	train	55.13	30.90	37.50	49.81	49.93	**59.43**	51.27
NanoClimateFever Retrieval	train	29.09	21.87	27.72	**44.57**	28.17	29.40	34.37
NanoDBPediaRetrieval	train	57.42	40.98	41.53	**61.81**	49.16	53.60	59.45
NanoFEVERRetrieval	train	88.59	56.74	57.84	**91.93**	80.95	82.60	90.54
NanoFiQA2018 Retrieval	train	46.25	17.92	26.05	53.61	**53.80**	53.22	53.41
NanoHotpot QARetrieval	train	75.15	42.22	51.20	**75.65**	67.68	65.87	72.86
NanoMSMARCORetrieval	train	61.17	30.93	41.23	**66.53**	48.93	53.50	61.97
NanoNFCorpus Retrieval	train	34.05	16.33	20.63	**36.85**	29.81	31.01	32.01
NanoNQRetrieval	train	69.86	28.61	36.17	**71.94**	61.82	57.89	66.80
NanoQuoraRetrieval	train	94.66	92.18	89.85	76.55	94.22	93.80	**95.87**
NanoSCIDOCSRetrieval	train	38.47	21.36	28.36	40.42	**41.56**	41.27	39.07
NanoSciFactRetrieval	train	71.75	24.97	33.39	**79.08**	71.41	72.98	70.20
NanoTouche2020 Retrieval	train	50.02	30.00	36.17	**53.24**	45.07	48.62	52.28
NarrativeQARetrieval	test	33.45	11.31	12.58	**34.13**	19.83	19.63	23.51
PIQA	test	30.40	12.58	15.64	**32.02**	28.67	30.35	30.65
PublicHealthQA	test	79.51	56.80	64.25	83.56	81.30	**84.56**	84.12
Quail	test	**4.38**	1.51	1.85	4.20	4.11	4.01	3.33
QuoraRetrieval	test	87.88	77.12	80.47	61.68	87.14	86.98	**88.14**
QuoraRetrieval HardNegatives	test	87.88	77.08	80.41	63.39	87.39	87.24	**87.98**
RARbCode	test	38.02	0.21	1.77	**54.61**	42.50	41.59	35.27
RARbMath	test	53.29	14.87	18.83	**74.57**	61.83	62.81	59.99
SCIDOCS	test	20.23	9.87	10.87	19.91	20.51	**20.85**	18.93
SCIDOCS-NL	test	8.19	5.30	7.94	**17.16**	14.43	16.71	15.50
SIQA	test	1.87	0.81	0.76	0.79	1.99	2.27	**2.31**
SciFact	test	67.32	28.65	33.64	**72.56**	68.18	66.95	65.21
SpartQA	test	7.67	0.67	3.22	0.73	0.10	1.82	**8.51**
StackOverflowQA	test	82.28	14.30	19.87	**90.79**	81.33	86.46	84.43
StatcanDialogue DatasetRetrieval	test	26.33	14.73	15.49	**33.44**	4.26	11.41	26.09
TRECCOVID	test	71.58	37.48	45.05	**77.34**	51.51	58.65	76.69
TempReasonL1	test	**1.49**	0.27	0.44	0.60	1.12	1.19	1.19
TempReasonL2Context	test	9.15	5.95	7.88	**11.73**	6.85	6.78	7.82
TempReasonL2Fact	test	15.07	6.26	7.92	**19.70**	11.98	11.72	16.03
TempReasonL2Pure	test	0.95	0.12	0.23	0.50	0.96	**1.14**	1.03
TempReasonL3Context	test	8.73	6.78	7.79	**11.62**	5.36	5.70	6.81
TempReasonL3Fact	test	13.43	6.94	7.29	**18.35**	9.69	10.19	12.33
TempReasonL3Pure	test	**6.30**	3.62	3.86	5.45	3.65	4.70	4.50
TopiOCQA	validation	14.13	5.59	6.04	**19.18**	12.52	12.04	15.16
TopiOCQA HardNegatives	validation	14.02	6.17	6.98	**19.19**	13.23	12.58	15.03

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 12: Retrieval and STS evaluation on English MTEB (Muennighoff et al., 2023) (3/3)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Touche2020 Retrieval.v3	test	**57.00**	27.64	33.36	55.40	47.28	52.60	56.19
WikipediaRetrieval Multilingual	test	90.91	73.24	76.80	91.78	89.89	90.40	**91.85**
WinoGrande	test	**49.60**	45.80	44.84	19.58	22.21	28.12	43.78
XMarket	test	30.73	7.26	13.61	**34.79**	14.79	21.83	29.33
XQuADRetrieval	validation	94.73	75.69	79.50	**96.11**	92.72	92.97	94.79
Average		47.04	24.69	27.83	**48.34**	42.02	43.47	46.46
STS Spearman correlation on cosine similarity								
BIOSSES	test	83.75	65.77	60.63	**84.52**	83.53	83.14	82.90
SICK-R	test	78.95	73.73	75.82	77.25	78.97	78.72	**82.40**
STS12	test	73.52	70.67	73.33	**78.34**	73.43	74.52	76.71
STS13	test	83.24	73.13	76.43	**87.10**	80.22	82.29	79.92
STS14	test	78.67	69.76	72.23	**80.15**	75.11	76.88	77.50
STS15	test	**87.46**	80.87	80.90	87.24	84.99	85.57	86.43
STS16	test	83.77	73.88	77.03	83.83	83.71	82.71	**85.19**
STS17	test	**89.78**	82.77	83.66	88.33	87.89	86.93	87.92
STS22.v2	test	65.84	55.21	58.06	65.50	63.46	**68.19**	67.00
STSBenchmark	test	84.93	74.51	79.45	85.09	84.30	84.46	**86.86**
STSBenchmark MultilingualSTS	test	84.93	74.51	79.45	85.09	84.30	84.45	**86.86**
SemRel24STS	test	**81.36**	76.07	79.75	77.70	80.19	81.29	79.31
Average		81.35	72.57	74.73	**81.68**	80.01	80.76	81.58

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 13: Retrieval and STS evaluation on Chinese MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
BelebeleRetrieval	test	15.96	63.34	70.18	91.32	92.23	**92.47**	91.57
BelebeleRetrieval	test	17.04	62.98	68.77	**92.24**	91.50	91.16	90.81
CmedqaRetrieval	dev	1.81	5.09	7.42	**35.89**	32.48	31.98	31.73
CovidRetrieval	dev	1.37	21.33	23.15	**78.91**	66.57	68.60	72.18
DuRetrieval	dev	3.92	19.05	27.65	**83.11**	76.88	77.85	78.50
EcomRetrieval	dev	4.33	13.65	22.54	**60.68**	29.77	47.77	55.86
LeCaRDv2	test	23.50	21.92	30.11	**58.40**	55.44	54.00	50.64
MIRACLRetrieval HardNegatives	dev	0.34	16.15	18.26	**57.66**	30.84	40.22	55.62
MLQARetrieval	test	6.29	31.77	36.78	**64.72**	58.95	59.40	64.69
MMarcoRetrieval	dev	7.06	28.30	37.94	**79.66**	70.05	69.96	76.28
MedicalRetrieval	dev	1.32	7.73	11.96	**56.63**	52.92	51.84	51.15
MultiLongDoc Retrieval	test	0.54	3.32	3.51	**17.17**	10.79	11.37	7.78
NeuCLIR2022Retrieval HardNegatives	test	1.99	24.86	29.41	**54.55**	39.14	44.28	49.03
NeuCLIR2023Retrieval HardNegatives	test	2.50	24.44	30.14	**50.10**	42.90	43.89	49.84
PublicHealthQA	test	17.66	59.55	65.23	84.56	85.38	**86.30**	84.94
T2Retrieval	dev	2.77	21.45	28.94	**83.16**	72.88	76.19	77.76
VideoRetrieval	dev	6.14	17.77	24.01	**70.45**	30.10	42.45	59.71
XPQARetrieval	test	18.03	30.56	42.41	**69.54**	66.79	66.23	65.72
XQuADRetrieval	validation	28.53	69.93	74.71	93.83	92.34	92.77	**94.08**
mFollowIR InstructionRetrieval	test	0.61	-2.44	**6.03**	2.80	4.33	4.39	1.57
Average		8.09	27.04	32.96	**64.27**	55.11	57.66	60.47
STS Spearman correlation on cosine similarity								
AFQMC	validation	7.65	10.97	14.60	**38.85**	35.27	36.19	36.56
ATEC	test	14.16	13.99	21.44	**44.80**	42.25	42.87	43.59
BQ	test	22.52	28.15	33.65	47.27	46.78	45.86	**55.03**
LCQMC	test	20.96	40.26	52.80	74.47	73.56	74.16	**75.42**
PAWSX	test	8.29	10.63	13.00	15.26	14.20	**15.96**	15.51
QBQTC	test	17.95	22.54	22.54	**34.35**	30.09	32.60	32.74
STS22.v2	test	44.52	43.00	50.37	**72.65**	71.58	71.00	71.38
STSB	test	34.50	65.25	72.05	81.35	80.18	80.52	**82.70**
STSBenchmark MultilingualSTS	test	36.19	65.46	71.87	82.42	80.78	81.15	**82.75**
Average		22.97	33.36	39.15	54.60	52.74	53.37	**55.08**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 14: Retrieval and STS evaluation on Hindi MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
BelebeleRetrieval	test	3.61	63.82	69.57	**89.21**	86.52	87.21	86.21
BelebeleRetrieval	test	43.80	28.07	31.67	**66.80**	66.14	64.76	62.00
IndicQARetrieval	test	2.11	32.46	35.63	**67.26**	63.22	62.60	63.87
MIRACLRetrieval HardNegatives	dev	0.14	21.87	23.79	**56.47**	43.20	43.77	54.80
MLQARetrieval	test	1.82	34.81	40.69	**63.77**	59.08	58.41	62.40
MintakaRetrieval	test	1.89	19.62	23.41	24.42	23.87	24.72	**26.48**
MultiLongDoc Retrieval	test	1.49	8.68	10.52	25.40	26.93	**30.47**	23.68
WikipediaRetrieval Multilingual	test	3.82	54.61	57.96	**86.57**	83.89	82.75	85.09
XPQARetrieval	test	20.45	57.18	64.98	**78.26**	77.26	77.50	74.36
XQuADRetrieval	validation	8.21	70.70	74.00	**93.66**	90.16	89.68	91.90
Average		8.73	39.18	43.22	**65.18**	62.03	62.19	63.08
STS Spearman correlation on cosine similarity								
SemRel24STS	test	36.09	68.11	73.78	83.62	83.35	**84.30**	79.55
Average		36.09	68.11	73.78	83.62	83.35	**84.30**	79.55

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 15: Retrieval and STS evaluation on German MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
BelebeleRetrieval	test	54.13	39.84	59.47	92.99	93.20	**93.21**	92.08
GerDaLIR	test	2.14	0.14	0.41	**16.18**	9.06	10.57	2.22
GerDaLIRSmall	test	6.18	0.47	1.34	**36.66**	21.23	25.33	6.76
GermanDPR	test	49.26	35.74	45.63	**82.47**	80.26	81.00	80.52
GermanGovService Retrieval	test	45.22	37.40	47.70	**88.71**	83.07	82.75	87.18
GermanQuAD-Retrieval	test	62.67	47.35	61.08	**94.15**	90.77	91.66	92.80
LegalQuAD	test	14.97	4.06	11.18	**58.83**	41.33	46.25	32.87
MIRACLRetrieval HardNegatives	dev	19.37	13.07	20.59	**53.37**	39.41	42.67	51.61
MLQARetrieval	test	33.09	17.87	28.39	**67.50**	63.06	61.94	65.77
MintakaRetrieval	test	20.37	9.66	17.62	27.05	29.08	30.10	**31.55**
MultiLongDoc Retrieval	test	13.25	3.15	9.74	37.81	34.15	**38.18**	24.10
WikipediaRetrieval Multilingual	test	62.29	44.00	58.90	**89.34**	89.01	88.04	88.75
XMarket	test	8.78	4.86	7.76	**28.28**	12.12	18.33	16.54
XPQARetrieval	test	52.72	32.65	48.28	**84.68**	83.44	83.50	82.31
XQuADRetrieval	validation	62.08	49.82	66.27	**95.79**	94.18	93.79	94.90
Average		33.77	22.67	32.29	**63.59**	57.56	59.16	56.66
STS Spearman correlation on cosine similarity								
GermanSTSBenchmark	test	63.94	54.53	66.52	82.98	82.17	82.25	**84.82**
STS22.v2	test	22.42	10.39	27.08	59.16	59.22	**61.35**	58.30
STSBenchmark MultilingualSTS	test	63.78	53.82	66.77	83.83	83.14	83.33	**85.68**
Average		50.04	39.58	53.45	75.32	74.84	75.64	**76.27**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 16: Retrieval and STS evaluation on French MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
AlloprofRetrieval	test	22.90	12.36	18.70	**54.39**	47.52	51.75	51.24
BSARDRetrieval	test	25.68	19.37	36.49	63.96	64.86	**65.77**	58.11
BelebeleRetrieval	test	66.86	53.05	71.63	**93.86**	90.26	90.65	92.38
FQuADRetrieval	test	51.07	36.70	43.68	**74.33**	68.04	67.03	70.59
MIRACLRetrieval HardNegatives	dev	26.57	17.58	19.93	**55.22**	38.43	42.81	52.72
MintakaRetrieval	test	22.91	15.57	20.15	26.94	28.39	29.17	**30.79**
MultiLongDoc Retrieval	test	41.80	20.64	26.17	59.84	59.76	**66.51**	54.17
PublicHealthQA	test	64.54	51.62	64.67	**92.05**	88.13	90.71	91.75
StatcanDialogue DatasetRetrieval	test	2.31	5.09	11.42	**22.68**	4.63	10.65	17.13
SyntecRetrieval	test	64.17	44.47	63.52	**84.03**	77.33	78.95	79.01
XPQARetrieval	test	51.10	34.80	49.81	**77.68**	76.80	77.56	74.84
Average		39.99	28.30	38.74	**64.09**	58.56	61.05	61.16
STS Spearman correlation on cosine similarity								
SICKFr	test	67.42	63.22	68.86	76.51	77.38	77.02	**80.55**
STS22.v2	test	66.65	61.01	73.31	**83.48**	77.99	81.16	79.90
STSBenchmark MultilingualSTS	test	69.82	67.60	74.97	83.33	81.79	82.40	**84.65**
Average		67.97	63.94	72.38	81.11	79.06	80.19	**81.70**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 17: Retrieval and STS evaluation on Spanish MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
BelebeleRetrieval	test	62.63	52.86	69.01	**93.52**	91.81	91.42	92.36
MIRACLRetrieval HardNegatives	dev	27.05	16.71	22.04	**51.08**	43.17	45.33	50.94
MLQARetrieval	test	42.28	29.25	40.56	**68.62**	62.89	61.50	66.69
MintakaRetrieval	test	21.47	16.29	20.21	26.93	29.00	29.76	**31.28**
MultiLongDoc Retrieval	test	41.13	14.04	20.55	62.07	63.86	**67.68**	55.56
PublicHealthQA	test	53.89	46.09	61.15	83.09	83.61	**86.62**	83.75
SpanishPassage RetrievalS2P	test	20.02	18.03	28.32	**43.11**	35.95	38.94	41.72
SpanishPassage RetrievalS2S	test	38.77	37.06	48.75	69.71	70.12	70.79	**72.98**
XMarket	test	12.02	6.38	10.22	**26.69**	14.20	20.96	17.51
XPQARetrieval	test	44.80	29.19	42.81	**72.14**	71.04	71.36	69.10
XQuADRetrieval	validation	73.39	64.76	75.47	**95.67**	92.84	92.20	94.40
Average		39.77	30.06	39.92	**62.97**	59.86	61.51	61.48
STS Spearman correlation on cosine similarity								
STS17	test	77.09	77.23	83.26	86.93	**87.31**	86.99	87.29
STS22.v2	test	57.19	46.44	53.42	75.58	75.16	**76.48**	76.01
STSBenchmark MultilingualSTS	test	71.90	65.44	75.15	84.72	82.76	83.35	**85.90**
STSES	test	74.00	72.02	66.90	75.89	76.47	75.22	**78.34**
Average		70.05	65.28	69.68	80.78	80.42	80.51	**81.89**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 18: Retrieval and STS evaluation on Japanese MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
BelebeleRetrieval	test	17.69	64.01	69.81	**91.73**	91.12	90.84	90.23
JaGovFaqsRetrieval	test	8.85	34.49	40.39	71.92	**72.73**	72.10	70.31
JaQuADRetrieval	validation	8.62	29.17	32.74	**54.78**	51.98	53.23	52.43
JaqketRetrieval	test	0.09	19.51	26.41	**46.91**	25.71	26.54	32.70
MIRACLRetrieval HardNegatives	dev	0.82	17.52	18.26	**66.26**	47.37	50.65	57.92
MintakaRetrieval	test	5.38	15.54	17.89	20.68	24.43	24.77	**25.57**
MultiLongDoc Retrieval	test	0.49	8.79	10.37	38.40	28.15	**38.88**	33.15
NLPJournal AbsIntroRetrieval	test	22.23	36.99	37.98	**98.75**	96.25	97.28	97.78
NLPJournal TitleAbsRetrieval	test	17.13	49.47	56.92	94.73	95.53	95.25	**95.77**
NLPJournal TitleIntroRetrieval	test	4.95	19.20	23.94	**93.40**	87.58	90.32	91.21
XPQARetrieval	test	32.25	43.38	56.12	75.55	75.89	**76.56**	73.22
Average		10.77	30.73	35.53	**68.47**	63.34	65.13	65.48
STS Spearman correlation on cosine similarity								
JSICK	test	60.57	77.93	78.38	**81.38**	80.56	80.28	80.45
JSTS	validation	51.09	76.85	80.11	80.69	82.39	82.64	**83.57**
Average		55.83	77.39	79.25	81.03	81.48	81.46	**82.01**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 19: Retrieval and STS evaluation on Russian MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
BelebeleRetrieval	test	9.30	63.01	71.61	**93.11**	90.74	90.96	92.41
MIRACLRetrieval HardNegatives	dev	1.15	18.44	21.74	**65.29**	42.54	49.71	59.74
MultiLongDoc Retrieval	test	2.85	15.13	20.25	49.54	54.01	**58.75**	46.55
NeuCLIR2022Retrieval HardNegatives	test	0.97	29.00	36.59	**58.08**	47.32	47.36	52.48
NeuCLIR2023Retrieval HardNegatives	test	4.26	28.95	34.23	**52.91**	47.55	49.54	51.30
PublicHealthQA	test	16.70	55.78	60.23	84.49	85.86	**87.28**	84.79
RiaNewsRetrieval	test	0.86	26.68	37.02	79.17	73.18	79.61	**81.43**
RiaNewsRetrieval HardNegatives	test	0.78	30.50	39.94	81.09	74.08	80.77	**82.99**
RuBQRetrieval	test	2.54	27.09	29.74	**72.27**	60.34	62.11	69.12
XQuADRetrieval	validation	14.28	69.50	75.24	**95.26**	91.97	91.53	94.15
mFollowIR InstructionRetrieval	test	-1.07	-4.64	0.97	-0.20	0.60	**2.66**	-2.92
Average		4.78	32.68	38.87	**66.46**	60.74	63.66	64.73
STS Spearman correlation on cosine similarity								
RUParaPhraserSTS	test	40.46	51.46	49.95	**74.57**	69.62	71.27	74.14
RuSTSBenchmarkSTS	test	51.02	70.40	76.15	81.54	81.52	81.36	**83.67**
STS22.v2	test	16.95	38.21	48.97	**71.00**	69.81	70.75	69.92
STSBenchmark MultilingualSTS	test	50.63	70.13	76.28	81.53	81.53	81.36	**83.44**
Average		39.77	57.55	62.84	77.16	75.62	76.18	**77.79**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 20: Retrieval and STS evaluation on Polish MTEB (Muennighoff et al., 2023) (1/1)

Task	Split	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
Retrieval nDCG@10 [%]								
ArguAna-PL	test	7.46	10.03	22.07	38.52	43.92	**46.80**	40.45
BelebeleRetrieval	test	30.29	35.21	53.19	**92.45**	89.16	89.63	90.97
DBPedia-PL	test	13.01	9.41	13.34	**34.88**	24.07	24.83	30.45
DBPedia-PL HardNegatives	test	16.36	15.28	18.87	**38.80**	27.67	27.98	35.09
FiQA-PL	test	2.47	3.10	6.56	**38.85**	26.67	29.02	32.71
HotpotQA-PL	test	19.09	12.95	16.80	**60.29**	49.51	48.40	55.11
HotpotQA-PL HardNegatives	test	24.86	19.64	23.17	**62.17**	51.89	51.08	56.15
MSMARCO-PL	test	11.30	7.83	16.01	**64.78**	43.29	49.89	59.49
MSMARCO-PL HardNegatives	test	32.40	22.79	28.04	**66.86**	52.27	55.61	63.51
NFCorpus-PL	test	7.40	9.99	11.26	**31.39**	27.00	28.71	27.66
NQ-PL	test	3.25	4.29	7.43	**54.01**	29.31	30.78	43.29
NQ-PLHardNegatives	test	5.27	6.57	10.43	**56.20**	32.12	32.49	45.98
Quora-PL	test	52.30	48.28	58.88	53.57	77.74	79.20	**80.65**
Quora-PL HardNegatives	test	54.25	49.15	60.60	54.82	77.87	79.01	**80.57**
SCIDOCS-PL	test	3.14	4.38	6.60	**15.38**	11.96	14.35	13.73
SciFact-PL	test	8.82	18.42	23.21	**64.88**	61.69	60.32	57.68
TRECCOVID-PL	test	12.15	16.74	30.85	**71.73**	51.25	58.40	70.81
XPQARetrieval	test	23.76	15.67	26.19	53.77	54.00	**54.60**	52.33
Average		18.20	17.21	24.08	**52.96**	46.19	47.84	52.04
STS Spearman correlation on cosine similarity								
CDSC-R	test	78.79	76.12	85.32	**91.97**	90.95	91.20	91.16
SICK-R-PL	test	53.67	48.13	53.26	72.90	74.85	75.11	**78.52**
STS22.v2	test	20.31	16.65	30.84	49.36	45.12	**50.53**	45.73
STSBenchmark MultilingualSTS	test	57.05	49.94	61.56	82.05	80.90	81.04	**82.62**
Average		52.46	47.71	57.75	74.07	72.95	74.47	**74.51**

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 21: Retrieval and STS evaluation on Cross-lingual MTEB (Muennighoff et al., 2023) (1/2)

Task	Split	Languages	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
			Retrieval nDCG@10 [%]						
BelebeleRetrieval	test	de,en	66.96	52.96	69.05	92.38	92.11	92.36	**92.40**
BelebeleRetrieval	test	en,de	46.64	35.59	62.72	**91.80**	90.18	91.19	89.84
BelebeleRetrieval	test	fr,en	79.80	64.85	73.07	**92.75**	88.64	90.50	92.32
BelebeleRetrieval	test	en,fr	62.43	54.01	74.08	**92.16**	89.35	91.16	90.53
BelebeleRetrieval	test	hi,en	8.07	68.55	71.82	**91.48**	87.95	89.73	90.46
BelebeleRetrieval	test	en,hi	2.88	65.36	72.27	**87.67**	82.39	85.74	83.21
BelebeleRetrieval	test	hi,en	55.56	33.15	36.43	**70.16**	68.62	68.77	69.62
BelebeleRetrieval	test	en,hi	44.96	24.18	32.34	**57.53**	51.68	54.83	51.97
BelebeleRetrieval	test	jp,en	25.11	67.58	71.78	91.17	88.68	90.24	**91.24**
BelebeleRetrieval	test	en,jp	7.18	60.99	70.07	**89.49**	86.80	88.90	85.39
BelebeleRetrieval	test	pl,en	45.49	48.50	63.12	**92.22**	88.98	90.17	91.70
BelebeleRetrieval	test	en,pl	28.36	36.52	58.30	**90.63**	89.00	90.34	88.36
BelebeleRetrieval	test	ru,en	15.23	64.24	71.26	**91.70**	87.98	89.50	91.25
BelebeleRetrieval	test	en,ru	7.67	63.76	73.50	**91.82**	89.75	91.20	90.16
BelebeleRetrieval	test	es,en	74.95	64.49	72.92	**92.65**	90.23	90.84	91.79
BelebeleRetrieval	test	en,es	58.15	53.79	72.11	**91.33**	89.96	91.11	90.96
BelebeleRetrieval	test	zh,en	31.42	62.78	68.42	89.68	88.79	89.78	**89.86**
BelebeleRetrieval	test	en,zh	8.69	61.78	69.85	**89.43**	85.59	88.81	86.22
BelebeleRetrieval	test	zh,en	27.30	62.34	67.54	**89.52**	87.45	88.94	89.48
BelebeleRetrieval	test	en,zh	7.50	61.05	69.84	**89.92**	86.40	88.74	85.50
BelebeleRetrieval	test	hi,hi	6.41	22.82	31.82	**58.73**	52.19	57.18	51.35
BelebeleRetrieval	test	hi,hi	2.23	31.65	35.49	**67.89**	65.01	65.83	64.99
CUREv1	dentistry_and_oral_health	es,en	9.53	11.85	22.29	44.83	37.08	40.17	**47.88**
CUREv1	dermatology	es,en	21.74	12.79	18.25	50.08	36.11	41.21	**56.62**
CUREv1	gastroenterology	es,en	18.31	12.20	20.48	48.85	38.38	41.16	**51.63**
CUREv1	genetics	es,en	19.82	11.87	21.38	**51.52**	39.55	40.01	50.42
CUREv1	neuroscience_and_neurology	es,en	9.30	9.07	18.68	**41.36**	29.13	31.69	40.34
CUREv1	orthopedic_surgery	es,en	8.69	5.04	15.17	37.66	39.86	38.10	**43.00**
CUREv1	otorhinolaryngology	es,en	8.30	6.23	14.92	38.43	32.59	33.32	**44.43**
CUREv1	plastic_surgery	es,en	11.80	8.55	18.93	42.34	34.48	36.89	**45.23**
CUREv1	psychiatry_and_psychology	es,en	14.34	15.93	24.81	50.45	39.39	41.25	**50.48**
CUREv1	pulmonology	es,en	13.17	13.76	23.95	**48.60**	36.02	36.85	47.91
CUREv1	avg	es,en	13.50	10.73	19.89	45.41	36.26	38.06	**47.79**
CUREv1	dentistry_and_oral_health	fr,en	12.04	12.18	21.30	42.74	34.24	37.53	**48.03**
CUREv1	dermatology	fr,en	27.76	15.22	19.91	50.74	34.70	38.40	**56.08**
CUREv1	gastroenterology	fr,en	21.87	12.85	19.89	48.14	34.75	37.14	**50.20**
CUREv1	genetics	fr,en	26.69	13.79	22.69	**49.52**	33.82	34.83	47.66
CUREv1	neuroscience_and_neurology	fr,en	14.39	10.59	16.83	38.74	24.63	27.98	**39.30**
CUREv1	orthopedic_surgery	fr,en	14.62	6.92	12.95	35.61	35.70	36.68	**44.31**
CUREv1	otorhinolaryngology	fr,en	10.57	7.09	14.12	40.14	29.33	30.62	**44.08**
CUREv1	plastic_surgery	fr,en	20.62	9.15	19.19	42.27	33.00	34.33	**45.15**
CUREv1	psychiatry_and_psychology	fr,en	20.73	14.97	25.78	**49.75**	36.05	37.79	48.87
CUREv1	pulmonology	fr,en	16.56	12.42	22.73	**46.90**	32.95	33.96	45.83
CUREv1	avg	fr,en	18.58	11.52	19.54	44.46	32.92	34.93	**46.95**
CrossLingualSemantic DiscriminationWMT19	test	de,fr	31.64	45.69	70.74	83.98	89.75	89.34	**91.11**
CrossLingualSemantic DiscriminationWMT19	test	fr,de	30.14	38.42	73.18	81.67	89.41	89.00	**90.83**
CrossLingualSemantic DiscriminationWMT21	test	de,fr	37.51	53.97	72.56	81.64	**89.81**	88.24	89.81
CrossLingualSemantic DiscriminationWMT21	test	fr,de	39.98	53.41	77.38	83.09	84.21	84.77	**87.23**
MLQARetrieval	test	de,en	45.85	26.14	35.89	**66.97**	60.34	60.86	65.22
MLQARetrieval	test	de,es	34.32	27.08	39.11	**70.58**	65.32	65.55	69.11
MLQARetrieval	test	de,hi	2.54	31.57	42.82	**68.43**	62.34	63.67	65.91
MLQARetrieval	test	de,zh	6.91	26.13	35.68	**66.87**	59.62	62.15	64.21
MLQARetrieval	test	en,de	33.03	17.24	32.36	**69.30**	66.14	66.89	67.81
MLQARetrieval	test	en,es	40.98	28.19	40.57	**66.38**	62.79	63.06	64.67
MLQARetrieval	test	en,hi	2.10	35.97	43.03	**61.04**	57.01	58.68	59.43
MLQARetrieval	test	en,zh	6.43	28.74	35.76	**59.75**	56.45	58.25	58.11
MLQARetrieval	test	es,de	32.10	21.12	36.79	**73.89**	68.02	67.89	71.58
MLQARetrieval	test	es,en	50.20	34.70	42.96	**68.75**	61.07	61.07	67.18
MLQARetrieval	test	es,hi	2.38	40.29	49.28	**68.74**	62.01	62.47	66.13
MLQARetrieval	test	es,zh	6.91	32.84	41.74	**68.56**	62.00	63.32	66.16

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 22: Retrieval and STS evaluation on Cross-lingual MTEB (Muennighoff et al., 2023) (2/2)

Task	Split	Languages	jc-v1	nllb-b	nllb-l	je-v3	jc-v2-s1	jc-v2-s2	jc-v2
MLQARetrieval	test	hi,de	8.05	21.71	39.25	**73.69**	67.16	67.60	72.32
MLQARetrieval	test	hi,en	10.28	36.62	41.89	**65.76**	56.77	58.33	64.18
MLQARetrieval	test	hi,es	8.33	32.66	45.79	**70.49**	64.31	65.37	69.20
MLQARetrieval	test	hi,zh	2.62	34.75	42.37	**67.60**	61.29	62.03	65.39
MLQARetrieval	test	zh,de	14.69	18.84	33.67	**67.35**	59.91	63.49	66.37
MLQARetrieval	test	zh,en	18.78	29.21	36.59	**60.33**	48.72	53.52	59.50
MLQARetrieval	test	zh,es	14.23	26.81	39.87	**64.46**	57.42	59.66	63.39
MLQARetrieval	test	zh,hi	2.43	36.52	44.44	**61.93**	54.28	57.14	60.20
XPQARetrieval	test	en,de	13.45	9.98	28.71	**60.48**	58.75	60.24	60.28
XPQARetrieval	test	de,en	39.28	29.44	47.12	82.12	82.80	**82.95**	79.45
XPQARetrieval	test	en,es	20.41	14.32	30.48	**53.46**	49.64	52.31	53.17
XPQARetrieval	test	es,en	40.61	31.30	45.34	68.31	69.14	**70.26**	67.62
XPQARetrieval	test	en,fr	22.20	15.66	31.46	54.80	52.15	**56.24**	55.54
XPQARetrieval	test	fr,en	47.06	36.14	49.88	73.83	75.44	**76.56**	72.96
XPQARetrieval	test	en,hi	4.87	27.87	34.86	**41.10**	36.15	38.23	39.63
XPQARetrieval	test	hi,en	8.07	56.85	62.61	**75.41**	74.52	74.41	71.56
XPQARetrieval	test	en,jp	5.12	17.09	28.58	47.12	47.19	**49.49**	46.51
XPQARetrieval	test	jp,en	19.43	46.09	54.54	73.50	71.75	**73.87**	69.75
XPQARetrieval	test	en,pl	12.08	9.86	16.86	34.19	34.23	**35.93**	34.56
XPQARetrieval	test	pl,en	20.69	17.50	25.65	50.88	51.56	**52.74**	49.98
XPQARetrieval	test	en,zh	4.41	12.86	19.12	**37.43**	34.62	36.07	36.21
XPQARetrieval	test	zh,en	12.96	32.39	39.80	**64.52**	62.18	60.96	58.95
mFollowIRCrossLingual InstructionRetrieval	test	en,ru	1.04	**1.55**	1.24	0.22	0.83	-0.20	-2.13
mFollowIRCrossLingual InstructionRetrieval	test	en,zh	0.18	-1.27	-1.47	-0.69	**3.07**	2.71	-0.72
Average			23.43	36.24	47.43	**69.84**	66.67	68.03	68.78
STS Spearman correlation on cosine similarity									
IndicCrosslingualSTS	test	en,hi	-11.26	58.03	58.73	67.84	62.49	61.18	**72.51**
STS17	test	en,de	53.67	54.88	65.87	83.66	84.18	84.03	**85.89**
STS17	test	es,en	52.80	64.91	77.17	85.09	85.60	85.27	**87.28**
STS17	test	fr,en	57.46	66.04	76.21	83.77	83.51	83.47	**85.91**
STS22.v2	test	de,en	55.96	56.41	54.70	61.63	65.06	**65.86**	61.14
STS22.v2	test	es,en	69.92	56.79	65.06	81.80	81.14	**82.53**	81.48
STS22.v2	test	pl,en	62.61	58.90	67.59	76.82	77.60	**82.35**	78.00
STS22.v2	test	zh,en	39.82	57.88	60.61	**76.33**	73.93	74.16	70.47
STS22.v2	test	de,fr	39.37	42.33	44.69	55.67	53.96	**60.86**	60.39
STS22.v2	test	de,pl	25.87	17.52	27.62	**56.69**	39.79	42.93	47.27
STS22.v2	test	fr,pl	**84.52**	50.71	39.44	84.52	50.71	61.98	61.98
Average			48.25	53.13	57.97	**73.98**	68.91	71.33	72.03

Evaluated on all tasks included in MTEB version 1.34.7, except the following: `MrTidyRetrieval`, `BrightRetrieval`, `MSMARCOv2`, `NeuCLIR2022Retrieval`, `NeuCLIR2023Retrieval` and `MIRACLRetrieval`. These tasks were excluded either due to bugs in the evaluation code or excessive computation times.

Table 23: MRL (Kusupati et al., 2024) ablation study on the CLIP Benchmark Retrieval tasks

Dataset - Dimension	1024	768	512	256	128	64
Zero-shot Image Retrieval - Recall@5 [%]						
Flickr30K (Young et al., 2014)	89.84	**89.98**	89.70	89.20	87.22	81.60
MS COCO (Chen et al., 2015)	**68.35**	68.26	68.16	67.43	64.58	59.41
Zero-shot Text Retrieval - Recall@5 [%]						
Flickr30K (Young et al., 2014)	98.00	**98.10**	98.00	98.00	96.30	93.40
MS COCO (Chen et al., 2015)	**81.46**	81.10	81.10	80.70	78.66	73.00

Table 24: MRL (Kusupati et al., 2024) ablation study on Crossmodal-3600 (Thapliyal et al., 2022)

Language - Dimension	1024	768	512	256	128	64
Zero-shot Image Retrieval - Recall@5 [%]						
Average	81.43	**82.35**	82.31	81.75	78.17	72.52
ar	**73.56**	73.39	73.17	72.61	68.42	62.28
bn	**63.78**	63.67	63.64	62.39	57.58	49.58
da	**85.39**	85.31	84.67	84.53	81.69	75.69
de	91.25	91.28	**91.47**	**91.47**	88.75	84.44
el	75.03	75.08	**75.25**	74.69	72.00	66.81
en	75.83	75.97	**76.03**	75.72	74.03	69.67
es	83.64	83.67	**83.86**	83.14	81.00	76.19
fi	82.83	**83.03**	82.61	81.67	79.44	74.94
fr	**88.78**	88.75	88.53	88.28	86.92	82.14
hi	**55.25**	54.97	55.14	54.14	50.06	42.33
id	**84.22**	84.00	84.11	83.25	80.00	74.17
it	88.33	**88.53**	88.31	87.83	85.11	79.75
ja	**87.03**	**87.03**	86.89	86.33	82.92	75.39
ko	**78.81**	78.72	78.56	77.03	73.44	65.78
nl	**82.56**	**82.56**	82.31	82.25	79.00	73.25
no	**81.08**	80.94	80.64	80.14	76.94	71.56
pl	84.00	83.97	**84.06**	83.39	81.28	76.83
pt	82.42	**82.50**	82.00	81.58	79.36	72.97
ro	89.36	**89.39**	89.22	89.14	87.03	82.06
ru	88.97	**89.17**	89.08	88.69	87.00	82.17
sv	**78.06**	77.86	78.00	77.47	74.83	69.53
th	81.61	**81.64**	81.14	80.36	76.97	68.17
tr	81.31	**81.44**	81.22	80.69	77.39	71.33
uk	88.56	88.42	**88.61**	87.89	85.86	80.72
vi	86.64	**86.72**	86.69	85.92	83.17	77.86
zh	78.97	**79.06**	78.86	78.08	74.81	67.36
Zero-shot Text Retrieval - Recall@5 [%]						
Average	83.23	**83.26**	83.21	82.81	80.54	75.37
ar	**76.25**	76.17	76.17	75.19	73.61	68.25
bn	**69.00**	**69.00**	68.94	68.75	66.19	59.39
da	**88.53**	88.39	88.25	87.58	85.69	80.25
de	92.47	**92.56**	92.42	92.31	90.64	87.19
el	73.33	**73.50**	73.47	73.39	72.75	68.61
en	78.58	**78.61**	78.36	77.86	76.44	72.58
es	86.28	**86.39**	86.33	85.78	84.03	80.17
fi	84.19	**84.25**	84.00	83.58	82.08	77.81
fr	**90.89**	90.86	90.75	90.14	88.61	84.89
hi	**61.64**	**61.64**	61.44	61.44	58.86	52.81
id	**86.31**	86.25	86.19	86.14	83.92	80.19
it	**90.17**	90.08	**90.17**	89.83	87.78	83.75
ja	88.50	**88.64**	**88.64**	87.92	86.22	81.64
ko	**81.42**	**81.42**	81.31	80.53	78.31	73.03
nl	**82.47**	82.44	82.36	81.94	79.97	76.00
no	83.75	**83.81**	83.78	83.28	80.39	75.39
pl	84.61	**84.67**	84.47	83.81	82.36	78.56
pt	**83.94**	83.83	83.64	83.25	81.72	77.00
ro	**91.31**	91.14	91.22	91.03	89.44	85.67
ru	**90.64**	90.58	90.31	90.19	88.50	84.86
sv	**78.28**	78.22	78.06	77.50	75.97	71.42
th	**81.94**	81.89	**81.94**	81.28	79.25	73.61
tr	82.67	**82.72**	82.39	81.92	80.36	76.06
uk	**89.53**	**89.53**	89.31	88.81	88.00	84.69
vi	**88.06**	87.94	87.61	87.47	85.92	82.00
zh	**79.22**	79.06	78.83	78.50	76.11	70.22

Table 25: MRL (Kusupati et al., 2024) ablation study on XTD10 (Aggarwal & Kale, 2020; Rajendran et al., 2016)

Language - Dimension	1024	768	512	256	128	64
Zero-shot Image Retrieval - Recall@5 [%]						
Average	**84.87**	84.85	84.60	84.32	81.80	77.85
de	**85.70**	85.40	84.90	84.70	81.10	79.30
en	89.40	**89.60**	88.70	88.90	86.50	83.00
es	85.90	85.90	**86.00**	85.70	84.00	80.80
fr	85.10	84.70	84.80	**85.30**	83.30	79.10
it	85.80	85.50	**86.10**	85.80	83.70	81.10
ko	**82.10**	81.90	**82.10**	80.10	77.90	71.90
pl	86.50	**86.70**	86.40	86.30	84.80	80.40
ru	81.10	81.20	80.80	**81.40**	77.80	74.00
tr	83.70	**84.10**	83.50	83.10	80.00	75.80
zh	83.40	**83.50**	82.70	81.90	78.90	73.10
Zero-shot Text Retrieval - Recall@5 [%]						
Average	**86.03**	86.02	86.02	85.84	84.37	81.02
de	86.20	**86.40**	86.10	85.90	85.20	81.50
en	**90.50**	**90.50**	91.10	90.40	90.10	86.80
es	**87.00**	86.80	86.70	86.90	86.10	81.80
fr	**85.20**	84.90	**85.20**	85.00	84.20	81.70
it	**88.00**	**88.00**	87.70	87.60	85.70	83.20
ko	82.10	82.20	82.40	**82.80**	80.30	77.50
pl	88.50	**88.70**	88.60	88.30	85.90	83.60
ru	**83.10**	83.00	82.70	82.80	80.80	78.10
tr	85.60	85.60	**85.90**	**85.90**	84.50	79.80
zh	**84.10**	**84.10**	83.80	82.80	80.90	76.20

Table 26: MRL (Kusupati et al., 2024) ablation study on English Classic MTEB (Muennighoff et al., 2023) Retrieval and STS tasks

Dataset - Dimensions	1024	768	512	256	128	64
Retrieval - nDCG@10						
Average	**49.33**	49.32	49.19	48.67	46.37	40.66
FiQA2018	41.93	**42.04**	41.79	40.96	38.75	34.27
NFCorpus	32.89	**33.01**	32.72	32.32	30.20	25.69
SciFact	**65.34**	65.12	65.26	64.57	62.51	57.17
SCIDOCS	**18.90**	18.85	18.84	18.46	16.98	13.84
CQADupstackRetrieval	**41.20**	41.17	41.06	40.47	37.97	32.42
Touche2020	23.94	24.08	24.23	24.57	**25.82**	23.26
TRECCOVID	**76.73**	76.54	75.98	76.01	73.10	67.47
FEVER	84.96	**84.99**	84.83	84.37	81.40	69.26
HotpotQA	**60.14**	60.13	59.71	58.06	52.14	38.97
DBPedia	**37.44**	37.42	36.98	36.12	33.21	26.34
NQ	57.17	**57.19**	57.02	56.30	53.49	45.42
ClimateFEVER	30.12	30.15	**30.35**	30.00	26.08	20.89
MSMARCO	**37.47**	37.43	37.45	37.16	35.74	32.47
ArguAna	**43.55**	43.50	43.53	42.83	41.06	37.21
QuoraRetrieval	**88.14**	88.13	88.06	87.88	87.10	85.19
STS - Spearman correlation based on cosine similarity						
Average	**81.29**	81.27	81.26	81.24	80.78	79.56
STS12	76.72	76.72	76.85	76.97	**77.13**	76.44
STS13	79.90	79.89	79.98	**80.48**	80.00	78.69
STS14	77.49	77.51	77.57	**77.67**	77.10	75.50
STS15	86.42	86.43	86.46	**86.58**	86.21	84.58
STS16	**85.18**	85.17	85.15	85.10	84.52	84.03
STS17	87.87	**87.89**	87.74	87.31	86.91	85.36
STS22	**67.07**	67.04	67.01	66.95	66.66	66.71
BIOSSES	83.00	**83.03**	82.65	82.33	81.18	78.27
STSBenchmark	**86.87**	86.68	86.86	86.72	86.26	85.14
SICK-R	**82.38**	**82.38**	82.36	82.27	81.87	80.89